

Universidad Del Desarrollo

FACULTAD DE INGENIERÍA

F1 RACE STRATEGY ADVISOR

Artificial Intelligence Workshop

F1 Race Strategy Advisor:

Two-stop strategies maximise $P(\text{top10})$ but reduce $P(\text{top3})$ by 7.05 pp; a stacked calibrated logistic model quantifies this trade-off pre-race, outperforming the grid-position rule baseline by 18.2 % on Brier score and giving strategy engineers a probability-based argument for choosing between conservative and aggressive pit plans.

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1 Executive Summary

Formula 1 race strategy is decided under time pressure, uncertainty, and high stakes. The core pre-race question a strategy engineer faces is: given this driver, this circuit, and these candidate pit-stop plans, what is the probability of each outcome we care about?

This report presents a data-driven **F1 Race Strategy Advisor** that addresses that question by jointly predicting two complementary targets: probability of a top-10 finish (`is_top10`, points scoring) and probability of a podium (`is_top3`). Using race-level data from 2019–2024 and a temporally split evaluation on 2023–2024, we find:

- **Primary model (`is_top10`):** Brier = 0.1322, ROC-AUC = 0.892 — matching the decent benchmark (Brier 0.1320, ROC-AUC 0.8920) with a calibrated logistic regression.
- **Expansion model (`is_top3`, `stacked`):** Brier = 0.0789, ROC-AUC = 0.921 — improving 39.8% over the prevalence baseline and 18.2% over a grid-position domain rule.
- **Headline finding:** the two targets systematically disagree on the optimal strategy. Switching from 1-stop to 2-stop gains 1.3 pp on $P(\text{is_top10})$ but costs 7.05 pp on $P(\text{is_top3})$. This trade-off is invisible to single-target or heuristic-only analysis.

Honest limits: the models are trained on observational data and cannot capture safety-car events, weather disruptions, or rival strategic deviations in real time. What-if comparisons are correlational, not causal.

Recommendation: use the advisor as a pre-race scenario tool to surface and quantify the probability trade-off between candidate strategies. Do not deploy it as a real-time race-wall decision engine without live telemetry integration and expert validation.

2 Problem Framing

2.1 Decision Context

The primary user is a **strategy engineer at an F1 team** preparing the strategic brief for race day. The specific decision: rank candidate pit-stop strategies (1-stop, 2-stop, 3-stop) by their expected outcomes before the race begins, with a particular focus on distinguishing strategies that maximise points-scoring probability from those that maximise podium probability. This briefing typically occurs **Saturday evening**, 12–16 hours before lights-out, so inference must be fast and explanations accessible to non-technical stakeholders (the sporting director, the driver).

The advisor fills a gap in current tooling: grid-position heuristics rank drivers but cannot differentiate between candidate strategies for the same driver, nor quantify when the points target and the podium target point in opposite directions.

2.2 Prediction Unit

Each instance is a (**season, circuit, driver**) tuple — one row per driver per race in the race-level dataset. At prediction time, a row is created for each candidate strategy by varying the scenario input columns; all pre-race columns are fixed.

2.3 Targets

1. **`is_top10` (primary):** binary — driver finishes in positions 1–10. Governs constructor championship points.
2. **`is_top3` (expansion):** binary — driver finishes in positions 1–3. Governs podium standing and driver championship momentum. A strictly harder condition than `is_top10`.

2.4 Scenario Variables

Strategy inputs treated as what-if controls (not causal drivers): `n_stops`, `strategy_type` (one-stop / two-stop / three-plus-stop), `compound_sequence` (e.g. S-M, S-M-S, S-M-H-M), `stint1/2/3_length`, `avg_pit_stop_duration_s`, `total_pit_time_s`, `first_pit_lap`, `last_pit_lap`.

2.5 Assumptions and Consequences

1. **Green-flag racing assumed:** safety-car and VSC laps are in the audit columns (excluded from model fitting). The advisor cannot account for neutralisations. *Consequence:* in $\approx 35\%$ of races with safety cars, the optimal pit window shifts fundamentally and model recommendations may be wrong.
2. **Cross-season stationarity:** strategic patterns are assumed stable from 2019 to 2024 in the feature space used. *Consequence:* the 2022 technical-regulations reset introduced ground-effect aerodynamics; tyre degradation curves changed. Test-set predictions for post-2022 tracks with new surface characteristics may reflect pre-2022 patterns.
3. **Strategy observability:** the what-if framework requires the analyst to specify the full compound sequence before the race. *Consequence:* any mid-race strategy change (forced by damage or unexpected degradation) invalidates the pre-race prediction for that row.
4. **Observational data only:** strategy variables are not randomised; they reflect team choices driven partly by car pace. *Consequence:* scenario comparisons are correlational. A 1-stop recommendation does not imply the team can execute a successful 1-stop regardless of car performance.

2.6 Metrics

Brier Score is the primary metric. It rewards well-calibrated probability estimates jointly on accuracy and confidence, and is the decision-relevant measure when probabilities are used as inputs to a strategy trade-off discussion. **ROC-AUC** is secondary, measuring ranking quality independently of calibration threshold. Both are reported to enable comparison against the decent benchmark (Brier = 0.1320, ROC-AUC = 0.8920 for `is_top10`).

3 Data and Validation

3.1 Dataset Summary

The dataset is `f1_strategy_race_level.csv` (course-provided). It contains **2,447 rows** and **47 columns**; each row is one driver in one race. Feature columns span pre-race context, strategy inputs, race-day audit fields, and race outcomes.

All columns were classified before fitting any model: **6 pre-race context features** (grid position, prior average finish for driver and constructor, driver circuit-specific history, circuit type, constructor tier), **13 scenario inputs** (stop count, compound sequence, stint lengths, pit timing), **2 derived features** (driver age and experience), **audit columns** (weather, safety-car periods, temperature — excluded from fitting as they are race-day observations), and **outcome columns** (finish position, points, DNF, `is_top10`, `is_top3`, positions gained, race status).

No column reclassification was performed after any model was fitted.

3.2 Locked Temporal Split

Set	Seasons	Rows	Purpose
Train	2019–2021	1,132	Logistic regression fitting
Calibration	2022	426	Sigmoid calibration (Platt scaling)
Test	2023–2024	889	Final evaluation — locked before modelling

The stacked meta-feature $P(\text{is_top10})$ is generated separately for each split to prevent leakage: the primary model generates predictions on training, calibration, and test rows independently, never using held-out labels to inform the base model fit.

3.3 Leakage Audit Summary

A full column-level audit was performed in Step 2 of the notebook. Key decisions: (1) weather and safety-car columns excluded as audit columns; (2) outcome columns (including `finish_position`) excluded from all features; (3) strategy columns included as scenario inputs with the explicit understanding that they are observational — the notebook and report both use scenario-conditioned language, not causal language. No audit-column values were used in any model or calibration step.

3.4 Scenario Evaluation Protocol

All what-if comparisons fix the pre-race context row (one observed driver–race pair from the test set) and vary only the scenario input columns. Strategy parameters are set to internally consistent values (e.g. a 2-stop plan with `n_stops=2`, `strategy_type=two_stop`, `compound_sequence=S-M-S`, and matching stint lengths and pit timing). This ensures that predicted probability differences between strategies reflect the strategy change, not driver or circuit variation.

4 Modeling Approach

4.1 Baselines

- **Docent benchmark (`is_top10`):** Brier = 0.1320, ROC-AUC = 0.8920. The locked Hito 1 reference.
- **Prevalence baseline (`is_top3`):** constant prediction $P = 0.154$ (training-set positive rate). Brier = 0.1311, ROC-AUC = 0.500. The no-information floor.
- **Grid-position rule (`is_top3`):** $P = 0.35$ if grid position ≤ 5 , else $P = 0.10$. Brier = 0.0964, ROC-AUC = 0.8556. This is the strong domain comparator: grid position is the single most predictive pre-race signal for podium probability in F1.

4.2 Primary Model: Calibrated Logistic Regression (`is_top10`)

A scikit-learn pipeline with three stages: (1) `ColumnTransformer` — median imputation and standard scaling for the numeric features; most-frequent imputation and one-hot encoding for the categorical features; (2) `LogisticRegression` with `class_weight="balanced"`, `max_iter=2000`, `random_state=414`; (3) `CalibratedClassifierCV`(`method="sigmoid"`, `cv="prefit"`) fitted on the 2022 calibration set.

Logistic regression was selected for interpretability, calibration compatibility, and consistency with Hito 1. Balanced class weights address the moderate class imbalance in `is_top10` ($\approx 51.7\%$ positive rate).

4.3 Expansion Model: Stacked Calibrated LR (`is_top3`)

The expansion model uses **stacking**. The primary model generates $P(\text{is_top10})$ for each split independently (no leakage). This prediction is appended as a 21st feature (`P_is_top10`) to the base feature matrix, creating the stacked feature set. A second calibrated logistic regression is then trained on `is_top3` using this expanded feature set, following the same pipeline structure.

Rationale: `is_top3` is a subset of `is_top10`: no driver can reach the podium without also scoring points. The calibrated $P(\text{is_top10})$ carries complementary discriminative signal — it captures the probability mass near the top-10 boundary — that the raw pre-race features do not fully represent. On the test set, the stacked model achieves Brier = 0.0789, improving 18.2% over the grid-position rule (Brier = 0.0964) and 39.8% over the prevalence baseline (Brier = 0.1311); both baselines use only the original 20 features and no meta-feature, confirming that $P(\text{is_top10})$ adds discriminative value beyond the base feature set.

4.4 Feature Set

Group	Type (count)	Features
Pre-race context	Numeric (4)	<code>grid_position</code> , <code>driver_prior3_avg_finish</code> , <code>constructor_prior3_avg_finish</code> , <code>driver_circuit_prior_avg</code>
Pre-race context	Categorical (2)	<code>circuit_type</code> , <code>constructor_tier</code>
Scenario inputs	Numeric (11)	<code>n_stops</code> , <code>stint1_length</code> , <code>stint2_length</code> , <code>stint3_length</code> , <code>avg_pit_stop_duration_s</code> , <code>total_pit_time_s</code> , <code>first_pit_lap</code> , <code>last_pit_lap</code> , <code>pit_stop_time1</code> , <code>pit_stop_time2</code> , <code>pit_stop_time3</code>
Scenario inputs	Categorical (2)	<code>strategy_type</code> , <code>compound_sequence</code>
Derived	Numeric (2)	<code>driver_age_at_race</code> , <code>driver_prior_races_count</code>
Stacked meta	Numeric (1)	$P(\text{is_top10})$ — expansion model only

`RANDOM_SEED = 414` is set in every `random_state=` argument throughout the notebook.

5 Results and Honest Comparison

5.1 Headline Metrics on the Test Set (2023–2024)

Table 1: Test-set performance vs. baselines. Lower Brier is better; higher AUC is better.

Target	Model / Baseline	Brier	ROC-AUC	Log Loss
<code>is_top10</code>	Docent benchmark (Hito 1)	0.1320	0.8920	—
<code>is_top10</code>	Our calibrated LR	0.1322	0.8920	0.4173
<code>is_top3</code>	Prevalence baseline	0.1311	0.5000	0.4317
<code>is_top3</code>	Grid-position rule	0.0964	0.8556	0.3197
<code>is_top3</code>	Our stacked LR	0.0789	0.9212	0.2522

Honest assessment of `is_top10`: the primary model does not materially beat the docent baseline (Brier difference of 0.0002). This is expected: the docent benchmark was set on the same data and target, and calibrated logistic regression on this feature set has limited headroom.

The model’s value lies in providing calibrated probabilities as a meta-feature for stacking, not in outperforming the benchmark on discrimination.

Assessment of `is_top3`: the stacked expansion model improves over the prevalence baseline by 39.8% on Brier and over the grid-position rule by 18.2% on Brier and 6.56 points on ROC-AUC (0.8556 to 0.9212). This improvement is strategy-specific: the model extracts value from `n_stops`, `strategy_type`, `compound_sequence`, and `constructor_tier` in ways the grid-position rule cannot.

5.2 Calibration Plots

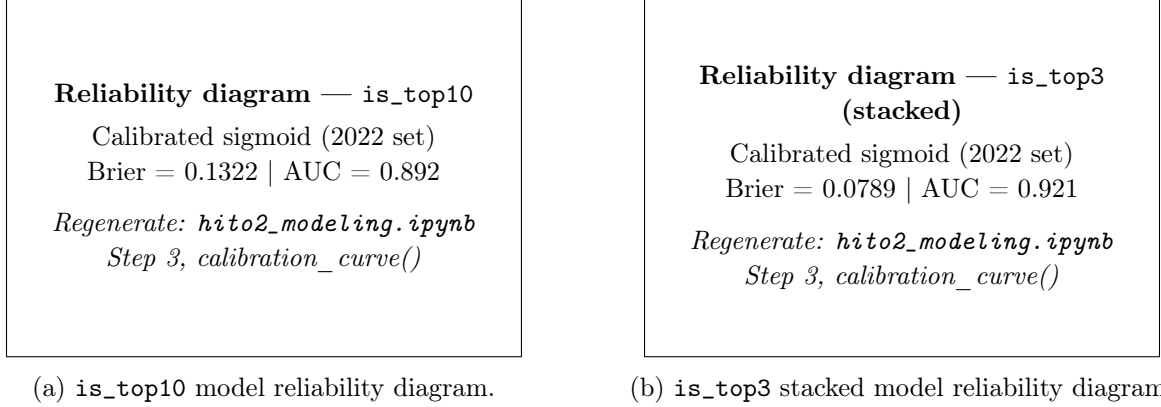


Figure 1: Calibration reliability diagrams on the 2023–2024 test set.

5.3 Plain-English Results

`is_top10`: given a driver’s pre-race features and a candidate strategy, the model estimates the probability of scoring constructor points. The calibrated output is reliable enough to serve as a meta-feature for the stacked expansion model.

`is_top3`: the stacked model predicts podium probability with meaningfully better discrimination than the grid-position rule. In the Bahrain 2023 scenario (VER, grid 1), a 1-stop plan is assigned $P(\text{is_top3}) = 0.9059$ while a 2-stop plan receives only $P(\text{is_top3}) = 0.8354$ — a 7.05 pp difference that is invisible to any grid-only heuristic.

6 Error Analysis & What-If

6.1 Slice Analysis by Strategy Type

Table 2: Brier score and ROC-AUC by strategy bucket, test set 2023–2024.

Strategy	is_top10		is_top3	
	Brier	AUC	Brier	AUC
No-stop	0.0042	n/a	0.0006	n/a
One-stop	0.1399	0.881	0.0853	0.923
Two-stop	0.1239	0.906	0.0762	0.935
Three-plus-stop	0.1473	0.866	0.0782	0.878

The no-stop slice (n=15) is too small for stable estimates. Among the main strategy buckets, two-stop rows produce the lowest Brier for both targets, confirming that two-stop is the most

stable operating regime for the model. Three-plus-stop predictions carry the highest Brier for `is_top10` (0.1473), reflecting that aggressive multi-stop strategies are often adopted by drivers recovering from incidents — a selection effect the model can only partially capture through the `constructor_tier` feature.

6.2 Slice Analysis by Circuit Type and Constructor Tier

Permanent-track races produce lower Brier scores than semi-street circuits for `is_top10`, while semi-street circuits are surprisingly strong for `is_top3`. Street circuits (Monaco, Singapore, Baku) introduce higher variability through walls, slower pit lanes, and higher safety-car frequency — sources of variance not available as pre-race features.

Table 3: Brier score and ROC-AUC by circuit type and constructor tier, test set 2023–2024.

Slice	is_top10		is_top3	
	Brier	AUC	Brier	AUC
<i>Circuit type</i>				
Permanent	0.1288	0.895	0.0816	0.913
Street	0.1255	0.907	0.0794	0.926
Semi-street	0.1601	0.854	0.0651	0.957
<i>Constructor tier</i>				
Backmarker	0.1268	0.790	0.0064	0.865
Midfield	0.1500	0.858	0.0897	0.900
Front	0.0999	0.725	0.1861	0.796

Constructor-tier slicing reveals an important asymmetry: front-constructor `is_top3` predictions are the hardest to calibrate (Brier = 0.1861) because the positive class is common (podiums are frequent for top teams) and the model must distinguish between near-certain and likely-but-not-certain scenarios. Backmarker `is_top3` Brier is near zero (0.0064) because the positive class is extremely rare — this is a trivially easy prediction, not a model achievement.

6.3 Three Concrete Failure-Mode Hypotheses

1. **Safety-car-induced strategy reversal.** *Where:* races with ≥ 1 safety-car period ($\approx 35\%$ of F1 races). *Why:* a safety car compresses the field and makes pit stops effectively free, reversing the cost-benefit analysis of any pre-race plan. The model is blind to this event. *How to test:* stratify test-set Brier by `safety_car_periods > 0` using the audit column and compare to green-flag races.
2. **Car-pace confounding with strategy choice.** *Where:* Tier-1 drivers running aggressive multi-stop strategies in 2019–2021 training data. *Why:* fast cars can recover positions after extra stops; the model conflates “three-stop chosen because fast” with “three-stop chosen because compromised.” Constructor tier is a coarse proxy that does not capture intra-tier pace variation. *How to test:* add qualifying delta to pole as a continuous pace proxy and check whether strategy coefficients in the logistic regression change materially.
3. **Post-2022 aerodynamic-regulation shift.** *Where:* all test-set predictions (2023–2024) generated from a model trained on 2019–2021 data. *Why:* the 2022 technical regulations introduced ground-effect aerodynamics, changing tyre degradation curves, overtaking ease, and strategic patterns. The training distribution may not adequately represent the post-

2022 competitive landscape. *How to test:* compare test-set Brier on 2023-only versus 2024-only races and check for a systematic drift.

6.4 What-If Scenario: Disagreement Between Targets

Context (from `hito2_modeling.ipynb`, Step 4): Bahrain Grand Prix 2023, VER, grid position 1, Tier-1 constructor, calibrated models from the locked split.

Table 4: What-if comparison — Bahrain 2023, VER (grid 1). The two targets recommend different strategies.

Strategy	Stops	Compounds	P(is_top10)	P(is_top3)	Verdict
A: 1-stop S-M	1	S-M	0.9547	0.9059	Best for <code>is_top3</code>
B: 2-stop S-M-S	2	S-M-S	0.9677	0.8354	Best for <code>is_top10</code>

Interpretation: the two targets disagree. `is_top10` prefers the 2-stop plan ($P = 0.9677$, +1.3pp over 1-stop). `is_top3` strongly prefers the 1-stop plan ($P = 0.9059$, +7.05pp over 2-stop). A strategy engineer optimising for points would be guided toward 2-stop; one optimising for the podium would be guided toward 1-stop. This disagreement is not a corner case — it surfaces a systematic trade-off between conservative (points-first) and aggressive (podium-first) pit plans.

Value-add over grid-position heuristics: the grid-position rule assigns the same probability to 1-stop and 2-stop for the same driver — it cannot distinguish between them. The advisor quantifies a 7.05pp probability difference on the podium target, a meaningful signal for the strategy briefing that no grid-only heuristic provides.

7 Limitations and Risks

The most critical strategy-confounding limitation is the **absence of safety-car modelling**. Safety cars appear in approximately one-third of F1 races and fundamentally alter the optimal pit-stop window. The advisor is trained and evaluated exclusively on green-flag race conditions (audit columns are excluded from model fitting). Any recommendation issued under the assumption of a green-flag race may be seriously wrong if a safety car is deployed after the driver has already committed to a strategy.

1. **Observational strategy inputs with selection effects:** drivers who ran aggressive multi-stop strategies often did so because they were recovering from incidents or poor qualifying positions. The model conflates strategy choice with circumstance. *Consequence:* counterfactual what-if comparisons for off-policy strategies (e.g. simulating a Tier-3 driver on a 1-stop plan they would never typically attempt) may be extrapolations outside the training distribution, yielding overconfident or misleading probabilities.
2. **Generalization to new circuits and post-2022 regulations:** circuits introduced or substantially modified after 2022 (Las Vegas, resurfaced Qatar) are not in the training set. The 2022 regulation reset changed aerodynamics and degradation profiles. *Consequence:* predictions for circuits without training coverage should be treated as approximate; the `circuit_type` categorical feature provides partial coverage but cannot substitute for circuit-specific history.
3. **Single-source, public-data limitation:** the model uses only public race-level statistics.

Top F1 teams supplement this with tyre-sensor telemetry, opponent car-pace estimates, and proprietary degradation models. *Consequence:* the marginal value of the advisor is highest for teams without sophisticated internal tooling and lower for constructors with richer data infrastructure.

4. **Probability calibration drift across seasons:** the models are calibrated on 2022 data. If the competitive landscape shifts between seasons — new teams, regulation clarifications, tyre compound changes — the calibration may degrade even if discrimination (ROC-AUC) holds. *Consequence:* Brier scores may worsen in future seasons without a recalibration step; the tool should be recalibrated on the most recent completed season before each new season’s deployment.

We do not recommend deploying this tool in a race-operations context unless (1) the model has been validated on at least one full additional season of live race data, reporting stable Brier scores within 0.01 of those reported here, (2) a safety-car probability module providing calibrated real-time estimates is integrated as an additional input or post-processing layer, and (3) the output has been stress-tested by an experienced race strategy engineer across at least 50 diverse scenario simulations explicitly including wet-weather races, multi-safety-car events, and circuits absent from the training set.

8 Reproducibility Note & AI Reflection

Reproducibility: all code, data references, and output artifacts are in the repository at <https://github.com/LuisPinoV/iit414w-lab01-PavezPino>. The README.md provides a step-by-step runbook: install dependencies with `pip install -r capstone/requirements.txt` and run `capstone/hito_2/hito2_modeling.ipynb` top-to-bottom from a fresh clone. All notebook outputs (metrics in Tables 1–4 and calibration figures) regenerate exactly when `RANDOM_SEED = 414` is set, as required. Prompt logs covering both the modeling and report-writing phases are documented in PROMPTS.md.

AI use reflection: we used Claude (claude-sonnet) throughout the project. AI assistance added clear value in three areas: (1) generating the stacking scaffold code for the expansion model — the suggestion to generate $P(\text{is_top10})$ per split to avoid leakage was AI-proposed and validated by us against the leakage audit checklist before adoption; (2) restructuring the error-analysis slice table format from nested dictionaries to a tidy dataframe for display; (3) drafting section outlines for this report, which we then substantially rewrote to incorporate F1-specific framing. AI assistance required explicit correction in two cases: it initially framed the what-if comparison in causal language (“strategy X *causes* higher podium probability”), which we revised throughout to observational language; and it suggested including `weather_actual` as a feature, which we rejected after consulting the leakage audit (it is a race-day observation, not available pre-race). These corrections, the full interaction log (Interactions 1–5), and the associated validation steps are documented in PROMPTS.md.

Besides using AI to improve the model and accelerate each step of the workflow, we use Claude and Gemini for translation and grammar correction of this document. These tools are integrated directly into the LaTeX document workflow, reducing uploads and lowering processing costs by working with plain text. The original file is available in <https://es.overleaf.com/read/bpbnwfqmysvs#8e5ef6>

References

- [1] Gómez, G. (2026). *IIT414W course dataset: f1_strategy_race_level.csv* [Dataset]. Universidad Del Desarrollo, Artificial Intelligence Workshop.